Lesson Notes

MODULE 4 | LESSON 3

THE THEOREM AND POLITICS OF EXTREME VALUES

Reading Time	25 minutes
Prior Knowledge	Extreme Value Theory, Central Limit Theorem
Keywords	Maximum Domain of Attraction

1. The Fisher-Tippett Theorem: The "Central Limit Theorem" of "Extreme Value Theory"

In previous readings, we have seen several references to the Fisher-Tippett Theorem, which as the name signifies is often attributed to Fisher and Tippett. A simply stated version of the extreme value theorem declares that "for a broad class of the distributions," as the threshold increases, the distribution of observations above that threshold converges to the Generalized Pareto Distribution (GPD).

The authors of our reading express a similar notion: "If the underlying distribution of the returns series belongs to the maximum domain of attraction (MDA) of the Generalised Extreme Value (GEV) distribution, as the threshold u becomes large, the distribution function of the exceedances over the threshold has approximately a "Generalized Pareto Distribution (GPD)" (Nortey et al.). However, this definition of the theorem begs the question, "What is an MDA?" to which we have the following answer:

A random variable belongs to the maximum domain of attraction (MDA) of a non-degenerate distribution G if, for suitable sequences, c_n and d_n (where c_n is greater than zero), the following holds:

$$\frac{M_n - d_n}{c_n} \xrightarrow{\mathcal{L}} G \quad \text{for} \quad n \to \infty.$$

(Quoted from Cízek et al.)

Alternatively, a CDF is said to be in the MDA of H "if there exist sequences of constants c_n and d_n greater than zero for all *n* such that

$$\lim_{n\to\infty} P\left(\frac{M_n - d_n}{c_n} \leq x\right) = H(x)$$

(Quoted from Haugh 7)

The point of showing this form of the MDA is to emphasize its similarity to the Central Limit Theorem (CLT). The Fisher-Tippett theorem has even been referred to as the "Central Limit Theorem of EVT" (Herman 5). As you recall, the CLT says "if you have a population with mean μ and standard deviation σ and take sufficiently large random samples from the population with replacement, then the distribution of the sample means will be approximately normally distributed." The analogous formulation is as follows: "Recall the CLT: If X_1, X_2, ... are IID with a finite variance then

$$\frac{S_n - a_n}{b_n} \xrightarrow{\mathcal{L}} N(0,1) \quad \text{for} \quad n \to \infty.$$

where

$$S_n := \sum_{i=1}^{n} X_i$$

$$a_n := nE[|X_1|]$$

$$b_n := \sqrt{n\mathrm{Var}[X_1]}$$

(Quoted from Haugh 6)

Of course, the CLT describes the distribution of that centrality measure, the mean; the FTT, on the other hand, is concerned with maxima. The other fundamental difference is the class of distributions, where according to CLT the means follow a Normal distribution and according to FTT, the maxima follow the GPD distribution (Herman 6).

If you have been reading astutely, you may be a little confused. In the lessons prior to this one, we have treated the GEV (and all three of its member distributions) as distinct from the GPD. They do not even have statistical approaches in common: Peaks over Threshold makes use of insights about the GPD, while the Block Maxima approach infers what it can from the Frechet, Gumbel, and/or Weibull distributions. And indeed, we should continue understanding them as separate, while we also appreciate their special underlying relationship, namely, that the “GEV distribution can be viewed as a special case of GPD under certain circumstances” (Ding 510). One can go so far as to say that the two distributions are “exchangeable” when the GPD threshold is high enough (Ding 509–10).

2. Political Risk

Nortey et al. note the significant impact of elections on stock market volatility. Indeed, political risk (sometimes referred to as “geopolitical risk”) is a global phenomenon affecting most every country (in so-called “developed” and “developing” countries alike) where a change in ruling party could mean “regime change” in both the political and statistical sense. It would be hard to overestimate the potential impact on the stock market of a new administration’s power to determine fiscal and (in many cases) monetary policy, decide budgets, issue audits, and pursue regulatory agendas.

Sun and Liu note five especially important political risk components:

- Government stability relates to, among other things, the frequency of elections and the (un)certainty of the outcome.
- Socioeconomic conditions includes such factors as GPD growth per capita, inflation and unemployment rates, and the GINI index.
- Investment profile can be indexed by the relative levels of imports and exports as well as the corporate tax rate.
- Internal conflict represents the crime rate (especially the murder rate), strikes, riots, and of course civil war.
- External conflict is exemplified by the Gulf Wars, the attacks on the United States on September 11, 2001, and even the Cuban Missile Crisis.

(Adapted from Sun and Liu 51)

In this vein, Agarwal and Agarwal use dummy variables to incorporate election vs. non-election years and stable vs. unstable governments in their GARCH(1,1) models to find strong evidence of both effects on the Indian stock market (484–6). Similarly, Wang and Lin embed dummy variables in their EGARCH(1,1) and GJR-GARCH(1,1) models to test whether the Taiwanese stock market performance is impacted by the democratization of Taiwan and/or the Taiwanese congress being in session. The reasoning that leads to the conclusions in both cases may be debatable (i.e., political), but these studies nevertheless demonstrate how the descendants of GARCH can be further extended and potentially improved.

Also worth noting here is that Wang and Lin observe that generally positive stock returns can be expected once any uncertainty has been clarified one way or the other (237). Of course, one can imagine any number of counter-examples to this generalization, but it nevertheless gets at the truth that markets abhor uncertainty (because people fear uncertainty, especially when it might impact their wealth), regardless of the market’s geography.

3. Conclusion

In this lesson, we applied a Peaks over Threshold approach to Value at Risk modeling using a GARCH model and fitting the residuals to a GPD. In the next lesson, we do something similar but instead of using just the standard GARCH(1,1) model, we use certain extensions of the model that add flexibility and/or incorporate some other new features.

References

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